Queue & Deque Problem Bank (Java)

Legend
★ = Core Placement Problem (Mandatory)
Covers internal implementation details alongside application-level problems.
These problems should be completed by every student before moving to Stack.

Section 1: Queue Basics and Creation

★ **Create a Queue using LinkedList**

★ **Create a Queue using ArrayDeque**

Create a Queue using PriorityQueue

Find Size of a Queue using size()

Section 2: Deque Basics and Creation

★ **Create a Deque using ArrayDeque**

Create a Deque using LinkedList

★ **Demonstrate Deque's Dual Nature as Both Stack and Queue**

Section 3: ArrayDeque vs PriorityQueue vs LinkedList Internals (Core Concept)

★ **Internal Structure of ArrayDeque (Resizable Circular Array)**

★ **Internal Structure of PriorityQueue (Binary Heap, Array Representation)**

★ **Internal Structure of LinkedList as a Queue (Doubly Linked Nodes)**

★ **Why ArrayDeque is Preferred over Stack/LinkedList for Queue Operations**

★ **Time Complexity Comparison: offer()/poll() — ArrayDeque vs PriorityQueue vs LinkedList**

★ **Heapify Process Demonstration in PriorityQueue**

Null Element Restriction in ArrayDeque and PriorityQueue

Section 4: Queue Operations

★ **Add Elements using offer()**

★ **Remove Elements using poll()**

★ **View Front Element using peek()**

Check if a Queue is Empty

Compare add() vs offer() (Exception vs Boolean Return Behavior)

Section 5: Deque Operations

★ **Add Elements at Both Ends using addFirst()/addLast()**

★ **Remove Elements from Both Ends using removeFirst()/removeLast()**

★ **View Elements at Both Ends using peekFirst()/peekLast()**

★ **Use Deque as a Stack (push/pop)**

Use Deque as a Queue (offer/poll)

Section 6: PriorityQueue and Custom Ordering

★ **Build a Min-Heap using Default PriorityQueue**

★ **Build a Max-Heap using Comparator.reverseOrder()**

★ **Custom Object Priority Queue using Comparator**

★ **Find K Largest Elements using PriorityQueue**

Find K Smallest Elements using PriorityQueue

Section 7: Queue/Deque-Based Logical Problems

★ **Reverse a Queue using a Deque**

★ **Check if a Sequence is a Palindrome using Deque**

★ **Implement a Circular Queue**

★ **Sliding Window Maximum using Deque**

Generate Binary Numbers from 1 to N using Queue

Section 8: Real World Queue/Deque Applications

★ **Print Queue Simulation (Task Scheduling)**

★ **Customer Service Ticketing System using Queue**

Browser History Navigation using Deque

★ **Task Priority Scheduler using PriorityQueue**

Undo/Redo Feature using Deque